

Lecture 16: Magnets

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 27. Feel free to read chapter 27 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

Answer the following questions about magnets. If the statement is false, rewrite it so that it becomes true.

- 1) (True or false) Magnets require only a (-) or (+) terminus.

False

- 2) (Fill in the blank) Fields created by magnets point from North to South or (+) to (-).

- 3) (True or false) Magnetic field lines use line density to represent field strength.

True

- 4) (fill in the blank) Magnetic field lines point North to the grade.

- 5) (Circle the correct answer) Ferromagnets/Electric currents have charges that are chemically locked into orientations.

- 6) (True or false) Currents moving through a wire produce an electric field that produces a magnetic field.

True

- 7) What are the units on magnetic field strength? Define these units in terms of MKS units.

Tesla, A/m

- 8) Write the conversion factor for Tesla to Gauss. How many Gauss equal 0.5 T?

$$500 \text{ Gauss} = 0.5 \text{ T}$$

- 9) Write the equation for force produced by electric currents in terms of the current, I , length of wire, l , and magnetic field B . Be sure to provide both the definition of force in terms of the cross product and trigonometric functions.

$$F = I (L \times B) = I L B \sin(\theta)$$